



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (CSE/ISE) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CS502 / 21IS502 - COMPUTER NETWORK AND COMMUNICATION

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I

1. a) Discuss different types of communication between two devices.
b) A nonperiodic composite signal has a bandwidth of 200 kHz, with a middle frequency of 140 kHz and a peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Approximately, draw the frequency domain of the signal.
2. a) Explain the working of Pulse-Code Modulation (PCM) for analog to digital signal conversion with a block diagram.
b) Explain different causes of transmission impairment.
3. a) Construct a codeword for a dataword 1 0 0 1 using a CRC encoder. Also, illustrate how the dataword is separated and accepted by the decoder. The shared divisor is 1 0 1 1.
b) Briefly explain link layer addressing.
4. a) Construct a frame using character and bit-oriented approaches and clearly illustrate stuffing and unstuffing in both cases.
b) Explain the working of CSMA/CD with the help of a flow chart.

Marks	BT*	CO*	PO*
8	L*2	1	1
4	L3	1	1,5
8	L2	1	1
8	L2	1	1
8	L3	2	2,5
4	L2	1	1
12	L3	2	2,5
8	L2	2	1

Unit – II

4. a) Discuss the factors that influence the performance of the network layer.
b) Explain IPv4 datagram with its header format.
c) Explain the concept of packet switching.
5. a) Explain the workings of unicast and multicast routing algorithms.
b) Explain the working of the link state routing algorithm and construct a database for the weighted graph shown in Fig. 5b.

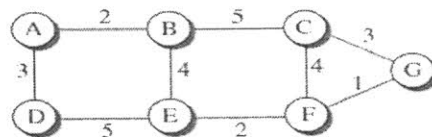


Fig. 5b

10	L3	4	1,5
----	----	---	-----

6. a) Construct a first distance vector for the internetwork shown in Fig. 6a and illustrate the updating after node B receives a copy (Fig. 6b) from node A.

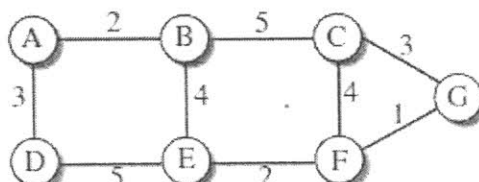


Fig. 6a

A	0
B	2
C	∞
D	3
E	∞
F	∞
G	∞

Fig. 6b

12	L3	4	1,5
----	----	---	-----

P.T.O.

- | | | | | | |
|----|--|---|----|---|---|
| b) | Explain the working of OSPF routing with an example. | 8 | L2 | 4 | 1 |
|----|--|---|----|---|---|

Unit – III

- | | | | | | | |
|----|----|---|----|----|---|-----|
| 7. | a) | Explain the connection establishment process using three-way handshake signals. | 8 | L2 | 5 | 2 |
| | b) | Explain different fields of TCP header segment with its format. | 8 | L2 | 5 | 2,5 |
| | c) | Briefly explain transport layer services. | 4 | L2 | 5 | 2 |
| 8. | a) | Explain the traditional client-server paradigm with a diagram. | 10 | L2 | 5 | 1,5 |
| | b) | Construct and explain the lost segment scenario during TCP operation. | 10 | L3 | 5 | 2,5 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
